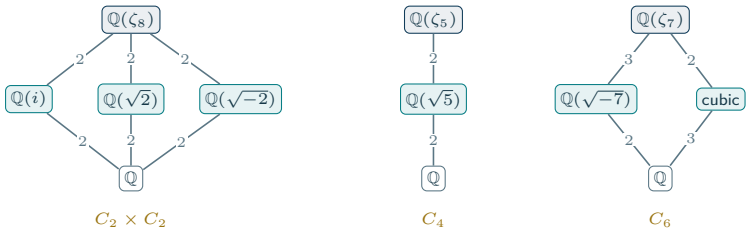
NUMBER THEORY ON ONE PAGE

Field extensions through the cyclotomic fields

Roots of unity give fields whose arithmetic is read from residues, and $n = 5, 7, 8$ show why $\sqrt{\pm 2}$ pair up.

1 / THE RULE AND THE PICTURE

An extension is a vector space, and its subfields mirror the subgroups of its Galois group



$C_2 \times C_2$ C_4 C_6

subfield lattices of three cyclotomic fields, edge numbers are degrees

Adjoining a root α of an irreducible polynomial f of degree d to K gives $K(\alpha) \cong K[x]/(f)$, a d-dimensional space with basis $1, \alpha, \ldots, \alpha^{d-1}$. Degrees multiply in towers:

$$[M : K] = [M : L]\,[L : K]$$

For $\mathbb{Q}(\sqrt{2}) = \mathbb{Q}[x]/(x^2 - 2)$ the degree is 2, and the automorphism $\sqrt{2} \mapsto -\sqrt{2}$ is its entire Galois group.

2 / THE CYCLOTOMIC FIELDS

$\mathbb{Q}(\zeta_n)$ has degree $\varphi(n)$ and Galois group $(\mathbb{Z}/n)^\times$, acting by $\zeta \mapsto \zeta^a$

n	$\Phi_n(x)$	$\varphi(n)$	$(\mathbb{Z}/n)^\times$	subfields
3	$x^2 + x + 1$	2	C_2	$\mathbb{Q}(\sqrt{-3})$, the Eisenstein field
4	$x^2 + 1$	2	C_2	$\mathbb{Q}(i)$, the Gaussian field
5	$x^4 + x^3 + x^2 + x + 1$	4	C_4	$\mathbb{Q}(\sqrt{5})$, with $\sqrt{5} = \zeta + \zeta^4 - \zeta^2 - \zeta^3$
7	$x^6 + x^5 + \cdots + 1$	6	C_6	$\mathbb{Q}(\sqrt{-7})$ and a real cubic from $\zeta + \zeta^{-1}$
8	$x^4 + 1$	4	$C_2 \times C_2$	$\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{-2})$

Here $\zeta_n = e^{2\pi i/n}$ and Φ_n is irreducible over $\mathbb{Q}$.
Each subgroup of $(\mathbb{Z}/n)^\times$ fixes a subfield, so the group tells us all subfields at once: the cyclic groups C_4 and C_6 give a chain and a diamond, while $C_2 \times C_2$ gives three fields side by side.

3 / HOW A PRIME FACTORS

For $p \nmid n$ the factorisation depends only on $p \bmod n$

n	prime p	$p \bmod n$	f	g	Φ_n modulo p
5	11	1	1	4	4 linear factors
5	19	4	2	2	2 quadratics
5	2, 3	2, 3	4	1	one quartic, inert
7	29	1	1	6	6 linear factors
7	13	6	2	3	3 quadratics
7	2	2	3	2	2 cubics
7	3, 5	3, 5	6	1	one sextic, inert
8	17	1	1	4	4 linear factors
8	3, 5, 7	3, 5, 7	2	2	2 quadratics, never inert

If f is the order of p modulo n, then p splits into $g = \varphi(n)/f$ primes, each of residue degree f, and Φ_n modulo p factors into g irreducibles of degree f. All of the table agrees with direct factorisation. The polynomial $x^4 + 1$ is irreducible over $\mathbb{Q}$ yet reducible modulo every prime below 2000, because $(\mathbb{Z}/8)^\times$ has no element of order 4.

4 / QUADRATIC SUBFIELDS AND THEIR SPLITTING RULES

Each is cut out by the residues of an index-two subgroup

field	conductor n	p splits when
$\mathbb{Q}(\sqrt{-3})$	3	$p \equiv 1 \pmod 3$
$\mathbb{Q}(i)$	4	$p \equiv 1 \pmod 4$
$\mathbb{Q}(\sqrt{5})$	5	$p \equiv \pm 1 \pmod 5$
$\mathbb{Q}(\sqrt{-7})$	7	$p \equiv 1, 2, 4 \pmod 7$
$\mathbb{Q}(\sqrt{2})$	8	$p \equiv \pm 1 \pmod 8$
$\mathbb{Q}(\sqrt{-2})$	8	$p \equiv 1, 3 \pmod 8$

A quadratic field $\mathbb{Q}(\sqrt{d})$ lies in $\mathbb{Q}(\zeta_n)$ exactly when its discriminant divides n, and a prime splits in it when the quadratic character $\left(\frac{d}{p}\right)$ is 1, which is a condition on $p \bmod n$.
The rules for $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$ are the primes mod 4 and mod 3 sheets, and the rows for $\mathbb{Q}(\sqrt{\pm 2})$ are the pair of the next sheet.

5 / CYCLIC AGAINST KLEIN

One quadratic subfield has no partner, three have each other

Because $(\mathbb{Z}/5)^\times \cong C_4$ and $(\mathbb{Z}/7)^\times \cong C_6$ are cyclic, each has a single quadratic subfield and a single quadratic character, so $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-7})$ stand alone and an inert prime is possible.
The group $(\mathbb{Z}/8)^\times \cong C_2 \times C_2$ has three quadratic characters whose product is trivial, $\chi_{-2} = \chi_{-1}\chi_2$, so a prime splits in exactly one of $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{-2})$ or in all three, and no prime is inert in $\mathbb{Q}(\zeta_8)$.
That relation is the duality of $\sqrt{2}$ and $\sqrt{-2}$, made possible by the third field $\mathbb{Q}(i)$.

ASIDE / WHY CYCLOTOMIC

Every abelian extension of $\mathbb{Q}$ lies inside some $\mathbb{Q}(\zeta_n)$ (Kronecker and Weber), so these fields model all the others.

LINKS TO OTHER SHEETS

The next sheet takes $n = 8$ in detail; the Gaussian and Eisenstein sheets are $n = 4$ and $n = 3$.

Convention.
$\zeta_n = e^{2\pi i/n}$; φ is Euler's function; C_k the cyclic group of order k; f residue degree and g number of primes above p; the table was checked by factorising Φ_n modulo p.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames